Extended Data Fig. 8 | Lineage-tracing reference: scLT-seq

a, CCE views of Day 2.5 (1,756 cells) and Day 5 (675 and 755 cells in two replicates), restoring the original lineage-tracing design. b, Original clone-transition examples for clone IDs 4, 6, 7, 23 and 32 retained as historical artwork; the exact clone-ID membership table is unavailable, so these arrows were not reconstructed from the separately saved expected-target pairs. c, Clone-based reference streams and all 19 method configurations for 1,756 cells, using the original cell-type grouping and colors. d, Per-cell cosine similarity to the clone-based reference. Numbers below method names in the stream panels are the median per-cell reference-direction cosines, shown to three decimal places in the style of main Fig. 4a. Boxes show the median and interquartile range, with 1.5-IQR whiskers and outliers; methods are ordered by median. All 19 configurations use barcode-aligned cells and the same reference coordinates. Fourteen configurations come from the supplied historical H5AD results; SDEvelo uses the previous preprocessed-input run, and TopicVelo, biVI, DeepKINET and cell2fate use the aligned raw-count extension. These runs do not constitute a uniform-preprocessing ranking. TFvelo projections were restored from saved velo_hat / fit_scaling_y fits using the original graph-and-embedding workflow because the stored projection duplicated scVelo(sto); no model was retrained. Cosines were calculated from the displayed per-cell directions. Main Fig. 6 retains its previously selected run summaries and was not recomputed. TopicVelo has 1,724 valid cosines; its 32 zero-direction cells remain undefined and are excluded from its boxplot. All other configurations have 1,756 valid cosines.